PADHAKU
Science · Class 10 Science



CHAPTER 5

Life Processes

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

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❓ Questions · 5.1 What are Life Processes?

Q 1 **Why is diffusion insufficient to meet the oxygen requirements of multi-cellular organisms like humans?**

✔ ANSWER

In multi-cellular organisms like humans, most cells are deep inside the body and are not in direct contact with the air.

Diffusion is a very slow process over long distances (oxygen would take about 3 years to reach our toes by diffusion).

So diffusion alone cannot supply enough oxygen to all cells quickly. Humans need a respiratory system and blood (haemoglobin) to carry oxygen.

Q 2 **What criteria do we use to decide whether something is alive?**

✔ ANSWER

Something is alive if it shows **life processes** such as nutrition, respiration, transportation and excretion.

Visible movement (breathing, growth) is a common sign, but the basic criterion is **molecular movement** inside it, which keeps repairing and maintaining its body structures.

Q 3 **What are outside raw materials used for by an organism?**

✔ ANSWER

- As **food**: a source of energy for all life processes.
- For **growth**: raw materials to build the body and make proteins and other substances.
- **Oxygen**: to break down food and release energy (respiration).
- **Water** and minerals: for various body processes.

Q 4 **What processes would you consider essential for maintaining life?**

✔ ANSWER

Nutrition, respiration, transportation and excretion are essential for maintaining life.

⭐ Activities · 5.2 Nutrition

Activity 5.1 **Variegated leaf (money plant/croton) kept in dark for 3 days, then in sunlight for 6 hours; boiled in water, then in alcohol, then dipped in dilute iodine solution.**

What happens to the colour of the leaf?

What is the colour of the solution?

What can you conclude about the presence of starch in various areas of the leaf?

✔ ANSWER

- In boiling alcohol the leaf loses its green colour and becomes pale/whitish. The alcohol turns **green** because chlorophyll dissolves in it.
- With iodine, only the parts that were **green** earlier turn **blue-black**; the non-green (white/yellow) parts do not.

Conclusion: Starch is formed only in the green parts. So **chlorophyll is essential for photosynthesis**.

Activity 5.2 **Two de-starched potted plants under air-tight bell-jars in sunlight for 2 hours; one jar has a watch-glass of potassium hydroxide (absorbs CO_2). Leaves tested for starch.**

Do both the leaves show the presence of the same amount of starch? What can you conclude from this activity?

✔ ANSWER

- **No.** The leaf from the plant with potassium hydroxide (no CO_2) shows **no starch** (no blue-black colour). The leaf from the other plant turns blue-black (starch present).

Conclusion: Carbon dioxide is essential for photosynthesis.

? **Based on the two activities performed above, can we design an experiment to demonstrate that sunlight is essential for photosynthesis?**

💡 SAMPLE ANSWER

Yes.

- Keep a potted plant in a dark room for 3 days to remove starch.
- Cover part of a leaf on both sides with black paper strips. Keep the plant in sunlight for about 6 hours.
- Pluck the leaf, remove the paper, boil it in water and then in alcohol, and test it with iodine solution.

Result: The covered part does not turn blue-black; the uncovered part turns blue-black.

Conclusion: Starch is made only where light falls, so **sunlight is essential for photosynthesis**.

Activity 5.3 **Two test tubes A and B with 1 mL starch solution (1%); 1 mL saliva added to A only; after 20–30 minutes a few drops of dilute iodine added.**
In which test tube do you observe a colour change? What does this indicate about starch? What does this

tell us about the action of saliva on starch?

✔ ANSWER

- Test tube **B** (no saliva) turns **blue-black**: starch is present.
- Test tube **A** (with saliva) shows **no blue-black colour** (stays the colour of iodine): starch is absent.

Conclusion: Saliva contains the enzyme **salivary amylase**, which breaks down starch into simple sugar.

? **What other function do you think is served by the acid (in the stomach)?**

💡 SAMPLE ANSWER

Hydrochloric acid also **kills germs (bacteria)** that enter the stomach with food. It also helps soften the food.

? **We have often heard adults complaining about 'acidity'. Can this be related to what has been discussed above?**

💡 SAMPLE ANSWER

Yes. 'Acidity' happens when the gastric glands release **too much hydrochloric acid**. The extra acid irritates the stomach lining and causes a burning feeling. Antacids (mild bases) are taken to neutralise it.

Questions · 5.2 Nutrition

Q 1 **What are the differences between autotrophic nutrition and heterotrophic nutrition?**

✔ ANSWER

Autotrophic nutrition	Heterotrophic nutrition
Organism makes its own food from simple inorganic substances (CO_2 and water).	Organism takes in complex food made by other organisms.
Needs sunlight and chlorophyll.	Does not need sunlight or chlorophyll.
Food is made by photosynthesis.	Food is broken down by enzymes.
Examples: green plants, some bacteria.	Examples: animals, fungi.

Q 2 **Where do plants get each of the raw materials required for photosynthesis?**

✔ ANSWER

- **Carbon dioxide**: from the air, through the stomata of leaves.
- **Water**: from the soil, absorbed by the roots.
- **Sunlight**: from the Sun, absorbed by chlorophyll.
- **Chlorophyll**: present in the chloroplasts of green parts (leaves).

- Minerals like nitrogen, phosphorus, iron and magnesium: from the soil.

Q 3 **What is the role of the acid in our stomach?**

✔ ANSWER

- Hydrochloric acid makes the medium in the stomach **acidic**, which helps the enzyme **pepsin** digest proteins.
- It also kills germs that enter with the food.

Q 4 **What is the function of digestive enzymes?**

✔ ANSWER

Digestive enzymes are biological catalysts that **break down complex food molecules into simple, small molecules** that can be absorbed.

E.g. amylase: starch → sugar; pepsin and trypsin: proteins; lipase: fats → fatty acids and glycerol.

Q 5 **How is the small intestine designed to absorb digested food?**

✔ ANSWER

- It is very **long and coiled**, giving a large area for absorption.
- Its inner lining has many finger-like projections called **villi**, which greatly increase the surface area.

- The villi are richly supplied with **blood vessels**, which take the absorbed food to every cell of the body.

⭐ Activities · 5.3 Respiration

Activity 5.4 **Air from a syringe/pichkari is passed through lime water in one test tube; exhaled air is blown through lime water in another.**

What does this tell us about the amount of carbon dioxide in the air that we breathe out?

✔ ANSWER

- Lime water turns **milky much faster** when we blow exhaled air into it than with air from the pichkari.

Conclusion: Exhaled air contains **much more carbon dioxide** than ordinary air.

Activity 5.5 **Fruit juice/sugar solution + yeast in a test tube with a cork and bent glass tube dipping into lime water.**

What change is observed in the lime water and how long does it take? What does this tell us about the products of fermentation?

💡 SAMPLE ANSWER

- The lime water slowly turns **milky** (after some time, depending on the temperature and amount of yeast).

Conclusion: Fermentation by yeast (anaerobic respiration) produces **carbon dioxide** (along with ethanol).

Activity 5.6 **Observe fish in an aquarium opening and closing their mouths and gill-slits. Are the timings of opening and closing of the mouth and gill-slits coordinated? Count how many times the fish opens and closes its mouth in a minute and compare with your breathing rate.**

💡 **SAMPLE ANSWER**

- **Yes**, they are coordinated: when the mouth opens, the gill-slits (operculum) close, and when the mouth closes, the gill-slits open. So water enters by the mouth and goes out over the gills.
- The fish opens and closes its mouth **many more times per minute** than we breathe (we breathe about 15–18 times a minute).

Reason: Water has much less dissolved oxygen than air, so aquatic animals breathe much faster.

❓ Questions · 5.3 Respiration

Q 1 **What advantage over an aquatic organism does a terrestrial organism have with regard to obtaining oxygen for respiration?**

✔ ANSWER

A terrestrial organism takes oxygen from the **air**, which has much more oxygen than the oxygen dissolved in water.

So it does not need to breathe as fast as aquatic organisms and uses less effort to get enough oxygen.

Q 2 **What are the different ways in which glucose is oxidised to provide energy in various organisms?**

✔ ANSWER

First, in the **cytoplasm**, glucose (6-carbon) breaks down into **pyruvate** (3-carbon) + energy. Then pyruvate is broken down in one of three ways:

- **Without oxygen (anaerobic), in yeast:** pyruvate → ethanol + carbon dioxide + energy (fermentation).
- **With oxygen (aerobic), in mitochondria:** pyruvate → carbon dioxide + water + a lot of energy.

- **Lack of oxygen, in our muscle cells:** pyruvate → lactic acid + energy (causes cramps).

Q 3 **How is oxygen and carbon dioxide transported in human beings?**

✔ ANSWER

- **Oxygen:** taken up in the lungs by the respiratory pigment **haemoglobin**, present in red blood corpuscles, and carried to tissues that need oxygen.
- **Carbon dioxide:** it is more soluble in water than oxygen, so it is mostly carried **dissolved in the blood plasma** from tissues to the lungs.

Q 4 **How are the lungs designed in human beings to maximise the area for exchange of gases?**

✔ ANSWER

- In the lungs the air passage divides into smaller and smaller tubes ending in millions of balloon-like **alveoli**.
- The alveoli give a very large surface (about 80 m^2 if spread out) for exchange of gases.
- Their walls are very thin and have an extensive network of **blood capillaries**.

⭐ Activities · 5.4 Transportation

Activity 5.7 **Find out the normal range of haemoglobin in human beings and in animals like buffalo or cow.**

Is it the same for children and adults? Difference between men and women? Is it different in calves, male and female animals? How would the difference be explained?

💡 SAMPLE ANSWER

- Normal haemoglobin (approx.): adult men about **13–17 g/dL**, adult women about **12–15 g/dL**, children about **11–13 g/dL**. Cow/buffalo: about **8–15 g/dL**.
- It is **not the same**: adults usually have more than children, and males more than females. In animals too, males usually have slightly higher levels than females, and calves differ from adults.

Explanation: Males usually have more muscle mass and higher oxygen needs; females lose blood in menstruation and need more iron in pregnancy. (Check exact values at the health centre.)

⭐ Activities · 5.4.2 Transportation in Plants

Activity 5.8 **Two pots with the same soil: one with a plant, the other with a stick of the**

same height. Soil covered with plastic, both sets covered with plastic sheets and kept in bright sunlight for half an hour.

Do you observe any difference in the two cases?

✔ ANSWER

- **Yes.** Water droplets appear on the inside of the plastic sheet covering the **plant**. No droplets form on the sheet covering the **stick**.

Conclusion: Plants lose water as vapour from their aerial parts (leaves) by **transpiration**.

❓ Questions · 5.4 Transportation

Q 1 **What are the components of the transport system in human beings? What are the functions of these components?**

✔ ANSWER

- **Heart:** muscular pump that pushes blood around the body.
- **Blood vessels:** arteries carry blood away from the heart; veins bring blood back to the heart; capillaries allow exchange of materials with the cells.
- **Blood:** plasma carries food, CO_2 and nitrogenous wastes in dissolved form; red blood corpuscles carry oxygen; platelets help blood clot at injuries.

- **Lymph:** carries digested and absorbed fat from the intestine and drains extra fluid from tissue spaces back into the blood.

Q 2 **Why is it necessary to separate oxygenated and deoxygenated blood in mammals and birds?**

✔ ANSWER

Mammals and birds have **high energy needs**, because they constantly use energy to maintain their body temperature.

Separating oxygenated and deoxygenated blood stops them from mixing, so the body gets a **highly efficient supply of oxygen**.

Q 3 **What are the components of the transport system in highly organised plants?**

✔ ANSWER

- **Xylem:** moves water and minerals from the roots to the other parts of the plant.
- **Phloem:** transports food (products of photosynthesis) from the leaves to other parts of the plant.

Q 4 **How are water and minerals transported in plants?**

✔ ANSWER

- Water and minerals move through the **xylem** (vessels and tracheids), which forms a continuous channel from roots to leaves.
- Root cells take up ions from the soil; water then moves into the root, creating **root pressure** that pushes water up (more important at night).
- **Transpiration pull:** evaporation of water from the leaves through stomata creates a suction that pulls water up from the roots. This is the main force during the day.

Q 5 **How is food transported in plants?**

✔ ANSWER

- Food is transported by **translocation** through the **phloem**, in the sieve tubes with the help of companion cells, both upward and downward.
- Sucrose is loaded into the phloem using energy from **ATP**. This raises the osmotic pressure, so water moves in.
- The pressure moves the food to parts with less pressure (storage organs, fruits, seeds, growing buds), according to the plant's needs.

❓ Questions · 5.5 Excretion

Q 1 **Describe the structure and functioning of nephrons.**

✔ ANSWER

Structure: A nephron is the basic filtering unit of the kidney. It has a cup-shaped **Bowman's capsule** around a cluster of thin-walled blood capillaries (glomerulus), followed by a long **coiled tubule** that joins a collecting duct.

Functioning:

- Blood is filtered in the capillary cluster; the filtrate is collected by Bowman's capsule.
- As the filtrate flows along the tubule, useful substances like **glucose, amino acids, salts and most of the water are re-absorbed** into the blood.
- The remaining liquid with nitrogenous wastes (urea, uric acid) is **urine**, which goes to the ureter.

Q 2 **What are the methods used by plants to get rid of excretory products?**

✔ ANSWER

- Oxygen and CO_2 are removed through stomata; excess water by **transpiration**.
- Some wastes are stored in **cell vacuoles**.
- Some are stored in **leaves that fall off**.

- Some are stored as **resins and gums**, especially in old xylem.
- Some wastes are **excreted into the soil** around them.

Q 3 **How is the amount of urine produced regulated?**

✔ ANSWER

The amount of urine depends on how much water is **re-absorbed** in the kidney tubules.

This depends on (i) how much **excess water** there is in the body and (ii) how much **dissolved waste** has to be excreted. More water in the body → less re-absorption → more urine.

📝 Exercises

1 **The kidneys in human beings are a part of the system for**

(a) nutrition (b) respiration (c) excretion (d) transportation

✔ ANSWER

(c) excretion – kidneys filter nitrogenous wastes from the blood to make urine.

2 **The xylem in plants are responsible for**

(a) transport of water (b) transport of food (c) transport of amino acids (d) transport of oxygen

✔ ANSWER

(a) transport of water – xylem carries water and minerals from roots; food and amino acids go by phloem.

3 **The autotrophic mode of nutrition requires**

(a) carbon dioxide and water (b) chlorophyll (c) sunlight (d) all of the above

✔ ANSWER

(d) all of the above – photosynthesis needs CO_2, water, chlorophyll and sunlight.

4 **The breakdown of pyruvate to give carbon dioxide, water and energy takes place in**

(a) cytoplasm (b) mitochondria (c) chloroplast (d) nucleus

✔ ANSWER

(b) mitochondria – aerobic break-down of pyruvate takes place in the mitochondria.

5 **How are fats digested in our bodies? Where does this process take place?**

✔ ANSWER

Fats are digested in the **small intestine**.

- **Bile** from the liver makes the food alkaline, and bile salts break large fat globules into small globules (**emulsification**), so enzymes can act better.
- **Lipase** in pancreatic juice breaks down the emulsified fats.
- Enzymes of the **intestinal juice** finally convert fats into **fatty acids and glycerol**, which are absorbed.

6 **What is the role of saliva in the digestion of food?**

✔ ANSWER

- Saliva **wets the food**, making it soft so it passes smoothly through the food-pipe.
- It contains the enzyme **salivary amylase**, which breaks down **starch into simple sugar**.

7 **What are the necessary conditions for autotrophic nutrition and what are its by-products?**

✔ ANSWER

Necessary conditions: carbon dioxide, water, **sunlight** and **chlorophyll**.

$6CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$ (in sunlight, with chlorophyll)

By-products: oxygen (and water). Extra glucose is stored as starch.

8 **What are the differences between aerobic and anaerobic respiration? Name some organisms that use the anaerobic mode of respiration.**

✔ ANSWER

Aerobic respiration	Anaerobic respiration
Takes place in the presence of oxygen.	Takes place in the absence of oxygen.
Glucose is broken down completely into CO_2 and water.	Glucose is broken down partly into ethanol + CO_2 (yeast) or lactic acid (muscles).
Pyruvate is broken down in mitochondria.	Takes place in the cytoplasm.
Releases much more energy.	Releases much less energy.

Organisms using anaerobic respiration: yeast, some bacteria (e.g. lactic acid bacteria in curd), tape-worm. Our muscle cells also respire anaerobically when oxygen is short.

9 **How are the alveoli designed to maximise the exchange of gases?**

✔ ANSWER

- There are millions of balloon-like alveoli, which give a **very large surface area** (about 80 m^2).
- Their walls are **very thin**, so gases diffuse easily.
- Their walls have an **extensive network of blood capillaries**.
- The lungs always keep a **residual volume of air**, giving enough time for O_2 to be absorbed and CO_2 released.

10 **What would be the consequences of a deficiency of haemoglobin in our bodies?**

✔ **ANSWER**

Haemoglobin carries oxygen. With less haemoglobin, the blood carries **less oxygen** to the cells, so less energy is released.

The person feels **tired and weak**, looks pale, gets breathless quickly and cannot work well. This condition is called **anaemia**.

11 **Describe double circulation of blood in human beings. Why is it necessary?**

✔ **ANSWER**

In one complete cycle, blood passes **through the heart twice**:

- **Pulmonary circulation:** deoxygenated blood from the body → right atrium → right ventricle → lungs (oxygenated) → back to the left atrium.
- **Systemic circulation:** oxygenated blood → left atrium → left ventricle → pumped to all body parts → deoxygenated blood returns to the right atrium.

Why necessary: It keeps oxygenated and deoxygenated blood **separate**, giving a highly efficient supply of oxygen. Humans need this because of high energy needs, e.g. to maintain a constant body temperature.

12 **What are the differences between the transport of materials in xylem and phloem?**

✔ ANSWER

Xylem	Phloem
Transports water and minerals.	Transports food (products of photosynthesis), amino acids, etc.
Movement is from roots upward to leaves (one direction).	Movement is both upward and downward (translocation).
Happens by physical forces: root pressure and transpiration pull; no energy used.	Uses energy from ATP to load sucrose (osmotic pressure).

Through vessels and tracheids.	Through sieve tubes with companion cells.

13 **Compare the functioning of alveoli in the lungs and nephrons in the kidneys with respect to their structure and functioning.**

✔ ANSWER

Alveoli	Nephrons
Structure: tiny balloon-like sacs at the end of the air tubes in the lungs.	**Structure:** long coiled tubules with a cup-shaped Bowman's capsule at one end, in the kidneys.
Thin walls with a network of blood capillaries.	Bowman's capsule surrounds a cluster of thin-walled blood capillaries.
Give a large surface for exchange of gases.	Give a large surface for filtration and re-absorption.
Function: O_2 from air passes into blood and CO_2 from blood passes into the air (gaseous exchange).	**Function:** blood is filtered; useful substances are re-absorbed and nitrogenous wastes are removed as urine.

Similarity: both are the basic units of their organs, have very thin walls and many capillaries, and remove wastes (CO_2 / urea) from blood.

🎉 **Chapter 5 done! Revise the answers once more tomorrow.**